DISCOVER · wide-field surveys find $\mathcal{O}$(1000) lensed AGN

CHARACTERIZE

Rubin LSST

pixel 0.2″ · PSF 0.71–0.77″

i r g · 10-yr

Euclid

pixel 0.1–0.3″ · PSF 0.16–0.5″

VIS Y J H · wide

Roman

pixel 0.11″ · PSF 0.087–0.13″

F158 F129 F106 · HLWAS

Lazuli WCC

pixel 17 mas · PSF 33–62 mas

LZz LZr LZg · 4×300 s

0.4 0.6 1 1.5 2 μm

Quad z_d=0.89 z_s=2.74

Quad z_d=0.43 z_s=1.87

Double z_d=0.57 z_s=1.22

1″

IFS field

D G C A B

LAZULI IFS · one pointing, every image

0.4–1.7 μm · R ≈ 100–500 · 40 mas spaxels · 4 × 900 s

λ

1.14–1.61 μm continuum

lens rest 0.60–0.85 μm (old stars bright): lens G shows

1.05 μm: Mg II line (rest 0.280 μm)

continuum removed: quasar broad lines only

0.57–0.72 μm continuum

lens rest 0.30–0.38 μm (old stars faint): quasars dominate

Same broad lines at the same redshift in every image → lensed quasar confirmed

quasar rest-frame wavelength [μm]

0.12 0.15 0.2 0.3 0.4

Lyα C IV C III] Mg II Hγ

A B C D G

f_λ [10^{-18} erg s^{-1} cm^{-2} Å^{-1}]

10 1 3

below noise in 1 h

0.5 0.7 1.0 1.3 1.7

observed wavelength [μm]

Simulated with SLSim: COSMOS hosts, qsogen quasar SEDs, static microlensing. Lazuli: 3.06 m, Strehl > 0.8 at 633 nm; specs from Roy et al. 2026 (arXiv:2601.02556). Lazuli throughput, sky, noise and exposure times are notional. Lensed-AGN counts: Oguri & Marshall 2010.